HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and deployed **Python**-based NLP models for document analysis on **AWS Lambda**, processing an average of **1,500 documents** per hour.
- Engineered **text preprocessing** pipelines using **NLTK** and **spaCy**, including tokenization, stemming, and lemmatization, to prepare large text corpuses for model ingestion.
- Implemented **vector embeddings** and **vectorization techniques** (e.g., TF-IDF, Word2Vec) to represent textual data for downstream machine learning tasks.
- Optimized the performance of **NLP models** for document classification and information retrieval, reducing inference time by **20%** through parameter tuning and efficient data loading.
- Integrated **machine learning models** into production environments via **REST APIs**, ensuring seamless data flow and real-time analysis capabilities.
- Collaborated with data scientists and AI engineers to refine and scale **AI Architectures**, contributing to technical direction for document processing initiatives.
- Managed code versions and deployed containerized applications using **Git** and **Docker** on **AWS EC2** instances, supporting iterative development cycles.

SKILLS

AI/ML – NLP, Information Retrieval, AI Architectures, Document Analysis, Model Optimization, Text Preprocessing, Vector Embeddings, Vectorization Techniques, Natural Language Processing (NLP), Optical Character Recognition (OCR)

Programming Languages – Python, R

Libraries/Frameworks – NLTK, spaCy, scikit-learn, pandas, NumPy, Flask, FastAPI

Cloud Platforms – AWS (EC2, Lambda, S3, CloudWatch), Google Cloud Platform (GCP), Microsoft Azure

Tools & Technologies – Git, Docker, REST APIs, Microservices, Jenkins, Jira, Confluence, PostgreSQL, MySQL, Oracle

Data Science Concepts – Algorithmic Thinking, Rapid Prototyping, Model Evaluation, Model Fine-Tuning, Data Science

PROJECTS

Document Analysis Pipeline for Health Records

- Developed a **Python** pipeline to ingest and analyze unstructured health records, extracting key entities and conditions using **NLP** techniques.
- Implemented **text preprocessing** and **vectorization** to create feature sets for a classification model, achieving **85% accuracy** in identifying relevant patient notes.
- Deployed the pipeline on **AWS Lambda**, handling approximately **500 documents** daily and integrating with an internal data store via **REST APIs**.

Information Retrieval System for Technical Documents

- Built a search solution that uses **vector embeddings** and **AI-powered search** principles to index and retrieve information from a large corpus of technical whitepapers.
- Engineered a **FastAPI** service to provide search functionality, optimizing query response times to under **200ms** for complex queries.
- Integrated **Git** for version control and containerized the application using **Docker** for efficient deployment and scaling.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)